Four-state thermal-overlap classification: working proof

Research working draft

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1 A finite classification for every simple four-state slow mode

The triplet test uses local rate information. A replacement can instead use the full generator's slow eigenvector. The result below is necessary and sufficient for ordinary *asymptotic* cooling and heating effects throughout the working domain: four distinct energies, a positive finite bath temperature, a reversible irreducible generator, and a simple slow nonzero eigenvalue. It does not classify all transient crossings or a degenerate slow eigenspace. Computing the full slow mode is part of the test; this is not another local rate-only criterion.

1.1 The geometry of Gibbs derivatives

Shift the energies to $(0, a, b, c)$ with $0 < a < b < c$, without changing their gaps. Write $\pi_i(\beta) = e^{-\beta E_i}/Z(\beta)$, and let a prime always mean a derivative in the initial inverse temperature. Define

$$g = \pi_1' > 0, \qquad u = \frac{\pi_2'}{\pi_1'}, \qquad v = \frac{\pi_3'}{\pi_1'}.$$

For any fixed vector w, including the left slow mode,

$$\frac{d}{d\beta}(w^\top \pi) = g\Phi(\beta), \qquad \Phi = A + Bu + Cv, \quad (A, B, C) = (w_1 - w_4, w_2 - w_4, w_3 - w_4). \tag{1}$$

The identity follows from $\sum_i \pi_i' = 0$.

Lemma 1 (Strictly convex derivative curve)**.** *For finite real β, $u'(\beta) < 0$ and v, considered as a function of u, has strictly positive second derivative. For $\beta \geq 0$, put*

$$c_0 = ab(b-a), \quad c_1 = ac(c-a), \quad c_2 = bc(c-b), \quad \gamma = \frac{c-a}{c-b} > 1, \qquad x = e^{-(c-b)\beta}.$$

Then

$$\frac{dv}{du} = s(x) = \frac{c_2 x^\gamma - c_0}{c_0 + c_1 x}, \qquad s'(x) = \frac{c_0 c_2 \gamma x^{\gamma-1} + c_1 c_2 (\gamma - 1) x^\gamma + c_0 c_1}{(c_0 + c_1 x)^2} > 0. \tag{2}$$

Here $s(0) = -1$, while $s(1) > -1$ need not be positive.

Proof. For functions $h_1, \ldots, h_k$ let $W(h_1, \ldots, h_k)$ denote the determinant with derivative orders $0, \ldots, k-1$ in its rows. The following identity holds for any number n of distinct ordered energies, with $f_i = e^{-\beta E_i}$ and $Z = \sum_i f_i$:

$$W(\pi_1', \ldots, \pi_k') = \frac{(-1)^{k(k-1)/2}}{Z^{k+1}} \left(\prod_{i<j\leq k} (E_j - E_i) \right) \left(\prod_{i=1}^{k} f_i \right) \sum_{j=k+1}^{n} f_j \prod_{i=1}^{k} (E_j - E_i), \quad k < n. \tag{3}$$

Indeed, adjoining a constant column gives $W(\pi_1', \dots, \pi_k') = W(1, \pi_1, \dots, \pi_k)$. Multiplying all $k+1$ functions by Z multiplies their Wronskian by Z^{k+1}: the product-rule transformation of the derivative rows is triangular with diagonal Z. Expand the first column of $W(Z, f_1, \dots, f_k)$. Terms with index at most k vanish by repeated columns; every remaining term is an exponential Vandermonde determinant with the displayed sign.

For $k = 2$ the identity gives $W(g, gu) = g^2 u' < 0$. For $k = 3$ it gives $W(g, gu, gv) = g^3 (u')^3 d^2 v/du^2 < 0$, proving strict convexity. For completeness, a mixed two-column calculation gives

$$W(\pi_1', \pi_3') = \frac{f_1 f_3}{Z^3}\{ab(b-a)f_2 - bc(c-b)f_4\},$$

whereas $W(\pi_1', \pi_2') = -f_1 f_2\{ab(b-a)f_3 + ac(c-a)f_4\}/Z^3$. Their ratio is dv/du and simplifies to (2). Differentiation gives the strictly positive displayed numerator. □

1.2 The finite test and its equality cases

The endpoint values needed below are explicit:

$$u(0) = 1 - \frac{4a}{a+b+c}, \qquad v(0) = 1 - \frac{4b}{a+b+c},$$
$$u(\infty) = -1, \qquad v(\infty) = 0.$$

At the bath and any other finite temperature, stable formulas are

$$u = \frac{-a + (b-a)e^{-b\beta} + (c-a)e^{-c\beta}}{a + be^{-(b-a)\beta} + ce^{-(c-a)\beta}},$$

$$v = \frac{e^{-(b-a)\beta}\{-b + (a-b)e^{-a\beta} + (c-b)e^{-c\beta}\}}{a + be^{-(b-a)\beta} + ce^{-(c-a)\beta}}.$$

Theorem 1 (Necessary and sufficient finite test)**.** *Let w be a nonzero left eigenvector of the simple slow nonzero mode, with $w^\top \pi(\beta_b) = 0$. Form A, B, C, Φ as in (1). If $C \neq 0$ and $-1 < -B/C < s(1)$, define β_* by the unique solution*

$$s(x_*) = -B/C, \quad 0 < x_* < 1, \qquad \beta_* = -\frac{\log x_*}{c-b}.$$

Otherwise omit β_*. Form the two finite sets of real numbers

$$\mathcal{V}_c = \{\Phi(0), \Phi(\beta_b)\} \cup \{\Phi(\beta_*) : 0 < \beta_* < \beta_b\},$$
$$\mathcal{V}_h = \{\Phi(\beta_b), w_1 - w_2\} \cup \{\Phi(\beta_*) : \beta_b < \beta_* < \infty\}.$$

An ordinary asymptotic cooling effect exists if and only if $\min \mathcal{V}_c < 0 < \max \mathcal{V}_c$. An ordinary asymptotic heating effect exists if and only if $\min \mathcal{V}_h < 0 < \max \mathcal{V}_h$. All inequalities are strict. A zero extremum without opposite signs does not certify an effect. Endpoints $\beta = 0, \infty$ are used only as limits; every effect asserted by the theorem has witnesses at finite positive temperatures.

Proof. If $C = 0$, then $\Phi = A + Bu$ is monotone (or constant). Otherwise $\Phi' = u'(B + C\, s(x))$. Lemma 1 shows that this has at most one zero for $\beta > 0$, precisely the β_* just specified. If $-B/C$ equals either limiting slope, there is no interior zero. Thus the listed values give the infimum and supremum of Φ on each side of the bath, including its limiting endpoint. Since $g > 0$, they decide whether $b'(\beta)$ takes both positive and negative values on that open side.

It remains to justify the equivalence with a Mpemba ordering; a stationary point alone would not suffice. Write $b = w^\top \pi$, so $b(\beta_b) = 0$. If b' has only one sign on a side, b is monotone there and $|b|$ increases with distance from the bath. There is no anomalous pair. Conversely, suppose b' takes both signs. The strictly convex representation implies at most two zeros of b' in total, counting multiplicities, unless it vanishes identically; the latter would require w constant and is excluded. On the cooling side consider the last sign-changing zero before the bath. On the interval immediately to its right, b has sign opposite to b', because $b(\beta_b) = 0$ and there are no further sign changes. Immediately to its left b keeps that nonzero sign but b' changes sign. Consequently $\partial_\beta |b| > 0$ there, yielding a finite cooling witness. For heating, use the first sign-changing zero after the bath. Immediately to its left b has the same sign as b'; immediately to its right the derivative changes sign while b does not, so $\partial_\beta |b| < 0$. This yields a finite heating witness. Isolated tangencies do not affect this reasoning. If the bath itself is stationary, apply the same argument on the open side. Strict opposite signs at a limiting endpoint persist at nearby finite temperatures by continuity. □

The theorem replaces an unbounded scan of initial temperatures by endpoint evaluations and at most one scalar equation with a unique solution. Its scope is every generator in the stated simple-mode working domain, with no restriction on energy spacing, rate scales, or triplet branches. It is an exact mathematical criterion, not a claim that floating-point arithmetic decides every equality case. With rational commensurate energies in an exponential coordinate and algebraic mode data, polynomial root isolation can certify the test. General transcendental boundary cases require exact arguments or suitable enclosures.

Strong effects can also be classified with finitely many tests. The same construction isolates every zero of b': on either side of the optional β_*, Φ is strictly monotone unless $B = C = 0$. Opposite endpoint signs give one unique root; an exact zero at β_* is a tangency. There are at most two finite positive roots in total. On each side of the bath, insert these stationary temperatures between the bath and the limiting endpoint (0 or ∞). Evaluate b at the resulting ordered knots. Every pair of consecutive nonzero values of opposite sign gives exactly one strong preparation in that open interval, since b is strictly monotone there. Every stationary knot with $b = 0$, other than the bath, is itself a strong preparation, including a tangential zero. A zero at 0 or ∞ is excluded. Intervals adjacent to a zero knot contain no additional zero. This is a complete finite classification of nonbath strong preparations; their actual temperatures, if desired, are the unique roots on those monotone intervals. It preserves the distinction between an ordinary turning point and a strong zero. The numerical implementation supplied here currently implements the ordinary-effect test; this strong-effect extension is a mathematical root-isolation prescription.

1.3 A sharp limit on thermal complexity

For four states, Lemma 1 shows that b' has at most two zeros counting multiplicity: a line intersects a strictly convex graph at most twice, with tangency counting twice. More generally, (3) shows that $(\pi_1', \ldots, \pi_{n-1}')$ is an extended complete Chebyshev system. This means every nonzero linear combination of its first k functions has at most $k - 1$ zeros counting multiplicity. The implication from nonvanishing prefix Wronskians is the standard repeated weighted Rolle argument (see, for example, Gasull, Geyer and Mañosas, *A Chebyshev criterion with applications*, J. Differential Equations 269 (2020), Lemma 2.3): divide by the first nonvanishing function, differentiate, and repeat; the transformed prefix Wronskians remain nonzero. At each differentiation the number of zeros can decrease by at most one. Thus any nonconstant Gibbs expectation has at most $n - 2$

stationary points. This is a consequence of established Chebyshev-system theory; no new general Wronskian or zero-counting principle is claimed.

Likewise, $Zb = \sum_i w_i e^{-\beta E_i}$ has at most $n-1$ zeros counting multiplicity by the exponential Vandermonde argument. One is the bath, so there are at most $n-2$ nonbath strong preparations. These two bounds are sharp even for positive reversible generators with a simple slow mode. Indeed, prescribe $n-2$ distinct positive nonbath inverse temperatures. Together with the bath they impose $n-1$ homogeneous interpolation conditions on n coefficients w_i. The exponential Chebyshev property makes the nullspace one-dimensional and its nonzero solution has exactly those simple zeros and no others. Rolle's theorem forces $n-2$ stationary points between them, attaining the derivative bound as well.

To realize this w, let $D = \mathrm{diag}(\pi_b)$, $N = w^\top D w > 0$, and choose

$$M = -I + \pi_b \mathbf{1}^\top + \epsilon \frac{D w w^\top}{N}, \qquad 0 < \epsilon < \min\left\{1, \frac{N}{\max_i |w_i|^2}\right\}. \tag{4}$$

Its off-diagonal entries are $\pi_{bi}(1+\epsilon w_i w_j/N) > 0$; its column sums vanish and detailed balance holds. The eigenvalues are $0, -1+\epsilon, -1, \ldots, -1$, with w the left slow mode. This elementary spectral realization is used to show attainability, without claiming a new general generator-construction method. It need not obey the triplet premise.

In particular, four is the minimum number of distinct states allowing both ordinary cooling and ordinary heating effects at one fixed bath with a simple slow mode: each side needs a sign-changing stationary point, while three states allow at most one globally. Four states attain the bound. Take $E_i = (i-1)\log 2$, $\beta_b = 1$, and

$$w = (-3/32,\ 11/16,\ -3/2,\ 1), \qquad \sum_{i=1}^{4} w_i x^{i-1} = (x-1/4)(x-1/2)(x-3/4).$$

The two nonbath strong preparations are $\beta = \log(4/3)/\log 2 < 1$ and $\beta = 2 > 1$. Choosing $\epsilon = 1/100$ in (4) gives an entirely rational generator, with eigenvalues $0, -99/100, -1, -1$. Both effects follow from the two exact nonbath zeros. This example illustrates attainability, not a second triplet counterexample. Publication priority for these Gibbs-specific consequences requires a separate literature assessment.